

Notes: Levchenko (RES, 2007)
Institutional Quality and International Trade

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Trade data	3
3.2	Country-level institutional quality	3
3.3	Industry-level institutional dependence	3
3.4	Factor abundance and factor intensity controls	4
4	Theoretical model	4
4.1	Basic goods and preferences	4
4.2	Incomplete contracts and relationship specificity	4
4.3	Autarky consequences	5
4.4	Trade equilibrium and institutional comparative advantage	5
4.5	Welfare logic: why the South may lose	5
4.6	North's gains	6
4.7	Ricardian comparison	6
5	Empirical specification and identification	6
5.1	Import-share prediction	6
5.2	Baseline regression	6
5.3	Identification logic	7
6	Tables and figures	7
6.1	Figures 1 and 2: FPE set and pattern of production	7
6.2	Figures 3 and 4: South's gains and production outside FPE	8
6.3	Tables 1 and 2: Baseline empirical results and alternative specifications	9
6.4	Table 3 and Appendix Table A.1: Alternative samples and sector examples	10
6.5	Appendix Tables A.2 and A.3: Summary statistics	11
6.6	Appendix Tables A.4 and A.5: Correlations and country sample	11

7	Short summary	12
8	Conclusion	13
8.1	Core conclusion	13
8.2	Economic intuition	13
8.3	Why the welfare result is important	13
8.4	Interpretation of the empirical evidence	13
8.5	Limitations	13
8.6	Takeaway	13

1 Big picture

- **Question.** Can cross-country differences in institutional quality be a source of comparative advantage, and do countries with better institutions export relatively more in industries that rely more heavily on institutions?
- **Core idea.** Institutions are not treated simply as taxes or productivity shifters. The paper models institutions as the degree of contract completeness in production relationships subject to relationship-specific investment and holdup.
- **Main theoretical result.** When institutional differences generate comparative advantage through incomplete contracts, the welfare implications differ sharply from a simple Ricardian productivity view. The North gains from trade, while the South may lose because trade eliminates the high-rent institutionally dependent sector in the South.
- **Mechanism.** Bad institutions create distortions and segmented factor rewards. In the institutionally dependent sector, one factor earns rents. Trade reallocates the institutionally dependent sector toward the country with better institutions, moving those rents internationally.
- **Empirical test.** The paper uses U.S. import shares by four-digit SIC industry and source country. It asks whether countries with better rule of law capture larger U.S. import shares in industries that are more institutionally dependent.
- **Industry institutional dependence.** The main proxy is product complexity: industries using a more dispersed set of intermediate inputs are assumed to rely more on contracts and institutions.
- **Main empirical result.** The interaction between country institutional quality and industry institutional intensity is positive and statistically significant across specifications and samples.
- **Economic takeaway.** Institutions shape not only the level of income but also the composition of international trade. Better institutions are a source of comparative advantage in institutionally intensive sectors.

2 Incremental contribution of the paper

- **Trade theory contribution.** The paper brings the incomplete-contracts view of institutions into a general-equilibrium trade model. This is different from treating institutions as a standard productivity parameter.
- **Welfare contribution.** The paper shows that institutional comparative advantage can generate nonstandard welfare effects: the South may lose from trade, labour in the South may lose, and factor rewards may diverge rather than converge.
- **Rent reallocation logic.** The paper links trade gains and losses to the reallocation of sectors with factor rents. Superior institutions allow the North to specialize in the more desirable institutionally dependent industries.
- **Comparison with Ricardian view.** Under the Ricardian view, the South gains by escaping its own bad institutions through imports. Under the incomplete-contracts view, the South can lose because it also loses its high-paying sector. This contrast is central to the paper's conceptual contribution.

- **Empirical contribution.** The paper tests for the institutional content of trade using variation across countries and industries, augmenting Romalis’s factor-content-of-trade specification.
- **Measurement contribution.** It uses the Herfindahl index of intermediate input use, multiplied by -1 , as a proxy for institutional intensity. More dispersed intermediate input use implies more production relationships and thus greater reliance on contract enforcement.
- **Robustness contribution.** The empirical results survive alternative institutional intensity measures, controls for factor abundance, raw materials, distance, financial dependence, development, and alternative country/industry samples.

3 Data and measurement

3.1 Trade data

The empirical analysis uses 1998 U.S. imports disaggregated by four-digit SIC industry and source country. The raw data include 177 countries and 389 industries; after requiring the institutional and factor variables, the final baseline sample contains 116 countries and 389 industries.

The dependent variable is the normalized U.S. import share:

$$rel_share_{ic} = \frac{\text{country } c\text{'s share of U.S. imports in industry } i}{\text{country } c\text{'s average share of U.S. imports}}.$$

This normalization removes the direct effect of country size and overall trade closeness to the United States.

3.2 Country-level institutional quality

The main institutional variable is the rule-of-law index from Kaufmann, Kraay, and Mastruzzi. It is intended to capture contract enforcement, property rights, and predictability of the legal system. The paper averages the index over 1996, 1998, and 2000. Higher values indicate better institutions.

3.3 Industry-level institutional dependence

The main industry-level proxy for institutional dependence is the Herfindahl index of intermediate input use, computed from the 1992 U.S. Input-Output Use Table and multiplied by -1 . The reason for using -1 times the Herfindahl index is that a low concentration of intermediate inputs means production depends on many suppliers, hence more contracts and relationships.

The paper also uses alternative measures:

- share of the 20 largest intermediates in total intermediate inputs;
- Gini coefficient of intermediate input use;
- number of intermediate inputs;
- investment-output ratio.

3.4 Factor abundance and factor intensity controls

To distinguish institutional comparative advantage from standard Heckscher-Ohlin forces, the empirical model includes:

- industry capital intensity;
- industry skill intensity;
- country physical capital per worker;
- country human capital per worker.

Some robustness checks add raw-material intensity and raw-material endowment, bilateral distance from the United States, financial dependence interacted with financial development, and industry fixed effects interacted with GDP per capita.

4 Theoretical model

4.1 Basic goods and preferences

The model starts from a small Heckscher-Ohlin style environment with two factors, capital K and labour L , and three goods:

- a pure capital good, K -good;
- a pure labour good, L -good;
- a mixed good, M , produced with both factors and therefore institutionally dependent.

Consumers have Cobb-Douglas preferences:

$$U(C_K, C_L, C_M) = C_K^\alpha C_L^\beta C_M^\gamma, \quad \alpha + \beta + \gamma = 1.$$

The pure goods are linear in one factor each:

$$p_K a = r, \quad p_L b = w.$$

4.2 Incomplete contracts and relationship specificity

The mixed good requires one unit of labour and x units of capital to produce y units of output. A fraction ϕ of capital's investment becomes relationship-specific. A higher ϕ means worse institutions: contracts and property rights are more incomplete.

After investment, the surplus is:

$$s = p_M y - w - r(1 - \phi)x.$$

With Nash bargaining, capital enters only if its participation constraint holds:

$$p_M y \geq w + (1 + \phi)rx.$$

This condition is the key institutional friction. In a country with worse institutions, ϕ is higher, so producing the mixed good requires a higher price or lower factor rewards.

4.3 Autarky consequences

Bad institutions generate two effects:

1. **Inefficiency.** Higher ϕ reduces the equilibrium size of the mixed-good sector. It also depresses base factor rewards w and r .
2. **Factor-market segmentation.** Labour employed in the mixed sector earns rents:

$$w + \phi rx.$$

These are the “good jobs” in the economy.

This segmentation is central. Trade can now create losses if it causes a country to lose the sector that pays rents.

4.4 Trade equilibrium and institutional comparative advantage

There are two countries: North and South. The North has better institutions:

$$\phi^N < \phi^S.$$

Since the mixed-good price condition is easier to satisfy in the North,

$$w + (1 + \phi^N)rx < w + (1 + \phi^S)rx,$$

the North has comparative advantage in the institutionally dependent good. Under trade and within the factor-price-equalization set, only the North produces the mixed good.

4.5 Welfare logic: why the South may lose

For the South, the gains from institutional comparative advantage can be written as:

$$[(r^T - r^S)K^S + (w^T - w^S)L^S] - \phi^S r^S x E^S L^S.$$

The first term is positive: the South gains efficiency by importing the mixed good produced with better Northern institutions. The second term is negative: the South loses the rents previously earned by labour in the mixed-good sector.

Therefore:

- the South may lose from trade if rent losses dominate efficiency gains;
- labour in the South is especially vulnerable because it loses access to the high-paying sector;
- capital in the South tends to gain from trade;
- factor rewards may diverge rather than converge.

4.6 North's gains

The North gains because its mixed-good sector expands to serve world demand. The North captures more of the rent-paying institutionally dependent production. The North's gains from institutional comparative advantage are:

$$\phi^N r^N x E^N (L - L^S).$$

Even though better institutions raise welfare in both autarky and trade, the gains from trade can be larger when North's institutions are imperfect enough to generate rents in the mixed sector. This is a subtle distinction between level effects and gains-from-trade effects.

4.7 Ricardian comparison

The paper contrasts the incomplete-contracts approach with a Ricardian view in which bad institutions simply destroy a fraction of mixed-good output:

$$p_M(1 - \tau)y = w + xr.$$

In that view:

- there are no factor-market rents;
- all countries gain from trade;
- the South gains by no longer producing the institutionally dependent good with bad institutions.

This is almost the opposite of the incomplete-contracts model. The comparison shows why modelling institutions as distortions and contract incompleteness is not equivalent to modelling them as productivity differences.

5 Empirical specification and identification

5.1 Import-share prediction

The empirical section extends the model to many countries and varieties of the mixed good. With Armington differentiation and product complexity, the model delivers:

$$\ln(s_{Mk}^l) \approx (1 - \sigma)n \ln(1 + \phi_k) + D_{lk}.$$

Because $\sigma > 1$, a country with better institutions, lower ϕ_k , should have higher import shares in more institutionally dependent industries, higher n .

5.2 Baseline regression

The empirical regression is:

$$\begin{aligned} rel_share_{ic} = & \alpha + \beta_1(inst_dep_i \times inst_c) + \beta_2(skint3_i \times skill_c) \\ & + \beta_3(capint3_i \times capital_c) + \gamma_c + \delta_i + \varepsilon_{ic}. \end{aligned}$$

Here:

- i indexes four-digit SIC industries;
- c indexes source countries;
- $inst_dep_i$ is industry institutional dependence;
- $inst_c$ is country institutional quality;
- γ_c and δ_i are country and industry fixed effects.

The coefficient of interest is β_1 . A positive β_1 means that countries with better institutions have higher U.S. import shares in institutionally intensive industries.

5.3 Identification logic

The identifying variation is a difference-in-differences style cross-sectional pattern:

- compare high- and low-institution countries;
- compare high- and low-institutional-intensity industries;
- ask whether high-institution countries are disproportionately represented in high-institutional-intensity industries.

Country fixed effects absorb country size, geography, and average propensity to export to the United States. Industry fixed effects absorb industry-level import demand and average tradeability. The interaction term isolates the match between country institutional quality and industry institutional dependence.

6 Tables and figures

6.1 Figures 1 and 2: FPE set and pattern of production

Read:

- Figure 1 shows the factor-price-equalization set in the world factor endowment box.
- The North must have enough capital and labour to produce the integrated-equilibrium amount of the institutionally dependent mixed good.
- Figure 2 shows the pattern of production at point A. The North produces the entire mixed good, while the South produces only the pure capital and pure labour goods.

Economic message: Superior Northern institutions allow the North to specialize in the institutionally dependent sector. This is the model's institutional comparative advantage channel.

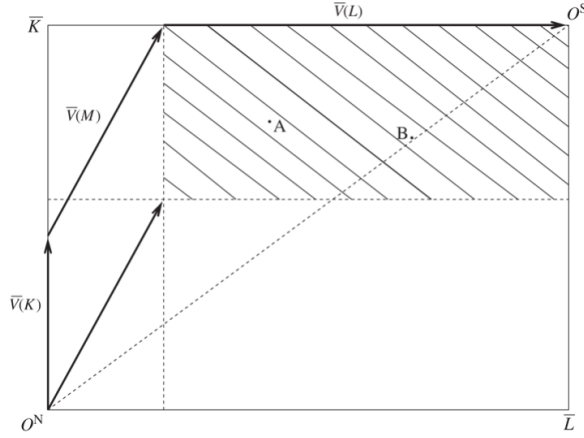


FIGURE 1

The world economy and the factor price equalization set

(a) Figure 1: World economy and factor price equalization set

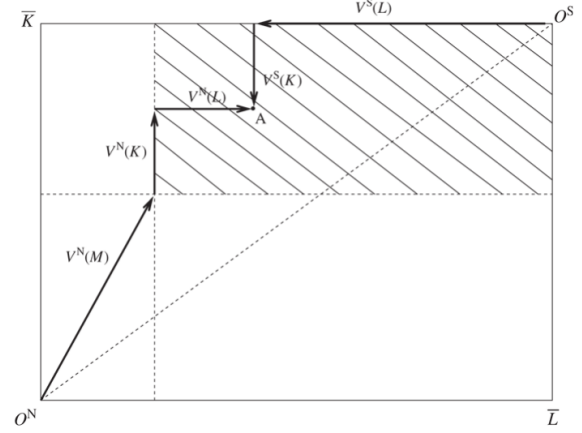


FIGURE 2

The pattern of production

(b) Figure 2: Pattern of production

6.2 Figures 3 and 4: South's gains and production outside FPE

Read:

- Figure 3 plots parameter regions where the South loses from trade and where labour in the South loses from trade.
- The horizontal axis is institutional weakness in the South, ϕ^S , and the vertical axis is institutional weakness in the North, ϕ^N .
- The area below the 45-degree line is relevant because the North has better institutions.
- Figure 4 shows a production pattern outside the FPE set. Even outside the clean FPE case, the key logic can remain: the mixed-good sector shrinks or disappears in the South.

Economic message: The South's welfare effect is ambiguous because it gains efficiency but loses rents. This is the main theoretical novelty relative to conventional comparative advantage models.

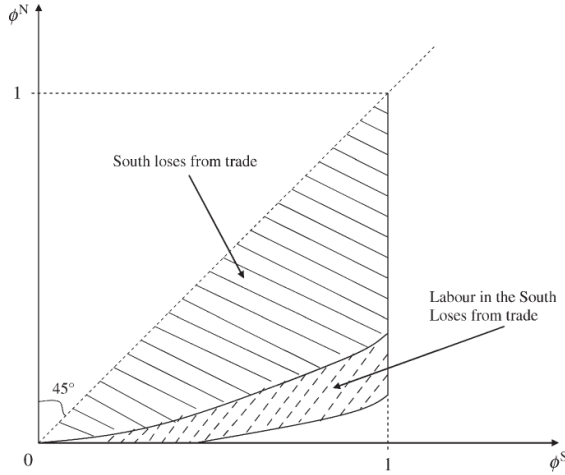


FIGURE 3

The South's gains from trade as a function of institutions

(a) Figure 3: South's gains from trade as a function of institutions

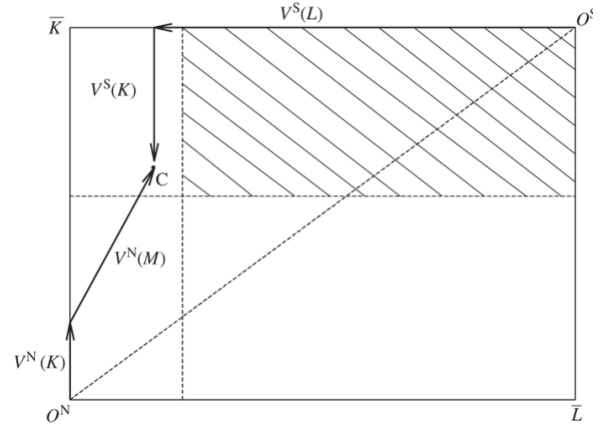


FIGURE 4

Production pattern outside FPE

(b) Figure 4: Production pattern outside FPE

6.3 Tables 1 and 2: Baseline empirical results and alternative specifications

Read:

- Table 1 tests whether countries with better institutions export more to the U.S. in institutionally dependent industries.
- Column 1 uses the Herfindahl-based institutional intensity measure. The coefficient on institutional intensity $\times$ institutional quality is 2.33 and significant at 1 percent.
- Columns 2-5 use alternative measures of institutional dependence. The interaction remains positive and statistically significant in every column.
- Table 2 adds tougher robustness checks: raw-material controls, distance interactions, financial development, cross-interactions with factor endowments, and sector dummies interacted with GDP per capita.
- Across all columns, the institutional interaction remains positive and significant.

Economic message: The empirical evidence supports an institutional content of trade. Better institutions predict larger import shares precisely in industries where institutions should matter most.

Dep. var: Normalized share of a country's imports in total imports	(1)	(2)	(3)	(4)	(5)
(Herfindahl index)*inst	2.33 (0.60)***				
(Share of 20 largest interm.)*inst		4.05 (0.53)***			
(Gini coefficient)*inst			20.70 (2.92)***		
(Number of intermediates/1000)*inst				3.12 (1.54)**	
(Investment/output)*inst					3.98 (1.47)***
(Skill intensity)*(skill endow)	11.35 (2.19)***	7.58 (2.22)***	7.66 (2.24)***	13.60 (2.12)***	14.11 (2.09)***
(Capital intensity)*(cap.endow)	0.50 (0.28)*	0.55 (0.28)*	0.54 (0.28)*	0.41 (0.29)	0.38 (0.28)
Country dummies	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes
Observations	31,568	31,568	31,568	31,568	31,568
Industries	389	389	389	389	389
Countries	116	116	116	116	116

Notes: Robust S.E. in parentheses. *Herfindahl index* of intermediate good use measures institutional intensity; *inst* is an index of institutional quality (rule of law) from Kaufmann *et al.* (2005); *capital intensity* = $1 - (\text{total compensation})/(\text{value added})$; *skill intensity* = $[(\text{nonproduction workers})/(\text{total employment})] * (1 - \text{capital intensity})$; *skill endow* and *cap. endow* are natural logs of human and physical capital per worker, respectively, obtained from Hall and Jones (1999). *Gini coefficient* of intermediate good use, *share of 20 largest intermediate*, *number of intermediates/1000*, and *investment/output* ratio are measure of institutional intensity. Variable definitions and sources are described in detailed in the text.

*Significant at 10%.

**Significant at 5%.

***Significant at 1%.

(a) Table 1: Baseline and alternative institutional intensity measures

p. var: Normalized share of a country's imports in total imports	(1)	(2)	(3)	(4)	(5)	(6)
erfindahl index)*inst	2.50 (0.92)***	2.00 (0.59)***	1.84 (0.82)**	2.36 (0.60)***	2.53 (0.83)***	3.64 (1.11)***
cill intensity)*(skill endow)	1.64 (3.11)	16.96 (3.21)***	1.43 (4.59)	11.21 (2.19)***	8.45 (2.75)***	4.25 (2.42)*
apital intensity)*(cap. endow)	0.72 (0.43)*	0.58 (0.28)**	0.12 (0.46)	0.51 (0.28)*	0.45 (0.37)	0.74 (0.46)
aw mat. intensity)*(raw endow)		40.10 (12.56)***	26.46 (14.72)*			
erfindahl index)*distance				-1.37 (0.75)*		
n. dep.)*(fin. development)					0.28 (0.08)***	
erfindahl index)*(skill endow)	-2.45 (4.52)		-2.61 (4.57)			
erfindahl index)*(cap. endow)	-0.04 (1.07)		0.12 (1.08)			
erfindahl index)*(raw endow)			-33.89 (17.82)*			
cill intensity)*inst	3.79 (0.89)***		6.37 (1.33)***			
apital intensity)*inst	-0.10 (0.59)		1.05 (0.78)			
aw mat. intensity)*inst			0.21 (0.70)			
country dummies	Yes	Yes	Yes	Yes	Yes	Yes
industry dummies	Yes	Yes	Yes	Yes	Yes	Yes
industry dummies*GDP per capita	No	No	No	No	No	Yes
observations	31,568	31,568	31,568	31,568	21,470	30,932
industries	389	389	389	389	376	389
countries	116	116	116	116	102	112

Notes: Robust S.E. in parentheses. *Herfindahl index* of intermediate good use measures institutional intensity; *inst* is an index of institutional quality (rule of law) from Kaufmann *et al.* (2005). In a three-factor model, *capital intensity* = $1 - (\text{total compensation})/(\text{value added})$; *skill intensity* = $[(\text{nonproduction workers})/(\text{total employment})] * (1 - \text{capital intensity})$. In a four-factor model, *raw material intensity* = $(\text{value of raw material inputs})/(\text{value of raw material inputs} + \text{value added})$; *capital intensity* = $1 - (\text{total compensation})/(\text{value added})$; *skill intensity* = $[(\text{nonproduction workers})/(\text{total employment})] * (1 - \text{capital intensity})$; *skill endow* and *cap. endow* are natural logs of human and physical capital per worker, respectively, obtained from Hall and Jones (1999). *Distance* is bilateral log distance between the U.S. and each trading partner, taken from CEPII. *Financial dependence* is a measure of dependence on external finance calculated from firm-level Compustat data following the methodology of Rajan and Zingales (1998). *Financial dependence*

(b) Table 2: Alternative specifications

6.4 Table 3 and Appendix Table A.1: Alternative samples and sector examples

Read:

- Table 3 tests whether the baseline result is driven by particular country groups or outlier industries.
- The institutional interaction remains positive and significant when the sample is restricted to the South, excludes Africa, excludes Southeast Asia, or drops the most institutionally intensive outliers.
- Table A.1 lists sectors with the lowest and highest institutional intensity.
- Low-intensity sectors include meat packing, soybean oil, poultry processing, and petroleum refining.
- High-intensity sectors include aircraft parts, surgical appliances, packaging machinery, wiring devices, and small arms ammunition.

Economic message: The result is not a North-South artifact or an outlier-sector artifact. The industry ranking also matches the intuition that complex manufactured goods rely more heavily on institutions.

Dep. var: Normalized share of a country's imports in total imports				
	(1)	(2)	(3)	(4)
(Herfindahl index)* inst	3.57 (1.20)***	1.56 (0.55)***	2.20 (0.65)***	2.28 (0.61)***
(Skill intensity)*(skill endow)	2.07 −(3.16)	18.78 (2.64)***	12.97 (2.17)***	11.52 (2.24)***
(Capital intensity)*(cap. endow)	0.42 −(0.41)	1.51 (0.26)***	0.55 (0.29)*	0.50 (0.29)*
Country dummies	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes
Observations	22,912	26,842	28,146	30,673
Specification	South only	No Africa	No SE Asia	No outliers
Industries	389	389	389	379
Countries	93	79	106	116

Notes: Robust S.E. in parentheses. *Herfindahl index* of intermediate good use measures institutional intensity; *inst* is an index of institutional quality (rule of law) from Kaufmann, *et al.* (2005); *capital intensity* = 1 − (total compensation)/(value added); *skill intensity* = [(nonproduction workers)/(total employment)]*(1 − capital intensity); *skill endow* and *cap. endow* are natural logs of human and physical capital per worker, respectively, obtained from Hall and Jones (1999). Variable definitions and sources described in detail in the text.
*Significant at 10%.

(a) Table 3: Alternative samples

Least institutionally intensive industries					Most institutionally intensive industries				
1	2011	Meat packing plants	1	3728	Aircraft parts and equipment, n.e.c.				
2	2075	Soybean oil mills	2	3296	Mineral wool				
3	2015	Poultry slaughtering and processing	3	3842	Surgical appliances and supplies				
4	2429	Special product sawmills, n.e.c.	4	3565	Packaging machinery				
5	2021	Creamery butter	5	3644	Noncurrent-carrying wiring devices				
6	2911	Petroleum refining	6	3643	Current-carrying wiring devices				
7	2026	Fluid milk	7	3482	Small arms ammunition				
8	2296	Tire cord and fabrics	8	3999	Manufacturing industries, n.e.c.				
9	2083	Malt	9	3321	Grey and ductile iron foundries				
10	2652	Setup paperboard boxes	10	2451	Mobile homes				

(b) Table A.1: Least and most institutionally intensive industries

6.5 Appendix Tables A.2 and A.3: Summary statistics

Read:

- Table A.2 summarizes industry-level variables. The Herfindahl-based institutional intensity measure has mean -0.133 , capital intensity has mean 0.623 , and skill intensity has mean 0.110 .
- Table A.3 summarizes country-level variables. The rule-of-law index ranges from -1.885 to 2.242 , showing large cross-country institutional variation.

Economic message: The empirical test has meaningful variation on both dimensions: industries differ in institutional dependence and countries differ in institutional quality.

	Mean	S.D.	Min	Max
Herfindahl index of intermediate use	−0.133	0.093	−0.780	−0.035
Capital intensity	0.623	0.114	0.181	0.951
Skill intensity	0.110	0.057	0.014	0.484

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(a) Table A.2: Industry-level summary statistics

	Mean	S.D.	Min	Max
Inst	0.094	1.052	−1.885	2.242
Log of human capital per worker	0.580	0.292	0.072	1.215
Log of physical capital per worker	9.203	1.587	5.763	11.589

(b) Table A.3: Country-level summary statistics

6.6 Appendix Tables A.4 and A.5: Correlations and country sample

Read:

- Table A.4 reports correlations between alternative institutional intensity indices.
- The Herfindahl measure is positively correlated with the share-20 and Gini measures, showing that the alternative proxies capture related but not identical dimensions of input complexity.
- Table A.5 lists the countries classified as North and South in the alternative-sample checks.

Economic message: The appendix clarifies that the empirical results are based on a broad country sample and are not dependent on a single measure of institutional intensity.

Correlation coefficients between alternative institutional intensity indices

	Share20	Gini	No. of int.	Inv/Out
Herfindahl index of intermediate use	0.67	0.7419	0.2314	0.1457

(a) Table A.4: Correlations between institutional intensity indices

Country	Country	Country	Country
Australia	Algeria	Guatemala	Paraguay
Austria	Angola	Guinea	Peru
Belgium	Argentina	Guinea-Bissau	Philippines
Canada	Bangladesh	Guyana	Poland
Denmark	Barbados	Haiti	Portugal
Finland	Benin	Honduras	Romania
France	Bolivia	Hungary	Russian Federation
Germany	Brazil	India	Rwanda
Hong Kong, China	Burkina Faso	Indonesia	Saudi Arabia
Iceland	Burundi	Jamaica	Senegal
Ireland	Cameroon	Jordan	Seychelles
Israel	Central African Republic	Kenya	Sierra Leone
Italy	Chad	Korea, Rep.	Somalia
Japan	Chile	Madagascar	South Africa
Netherlands	China	Malawi	Sri Lanka
New Zealand	Colombia	Malaysia	Sudan
Norway	Comoros	Mali	Suriname
Singapore	Congo, Dem. Rep.	Malta	Syrian Arab Republic
Spain	Congo, Rep.	Mauritania	Tanzania
Sweden	Costa Rica	Mauritius	Thailand
Switzerland	Cote d'Ivoire	Mexico	Togo
Taiwan, POC	Cyprus	Morocco	Trinidad and Tobago
U.K.	Dominican Republic	Mozambique	Tunisia
	Ecuador	Myanmar	Turkey
	Egypt, Arab Rep.	Nicaragua	Uganda
	El Salvador	Niger	Uruguay
	Fiji	Nigeria	Venezuela, RB
	Gabon	Oman	Yemen, Rep.
	Gambia, The	Pakistan	Yugoslavia, Fed. Rep.
	Ghana	Panama	Zambia
	Greece	Papua New Guinea	Zimbabwe

(b) Table A.5: Country list

7 Short summary

- The paper studies how institutional quality affects international trade patterns.
- It models institutions as contract completeness in an incomplete-contracts framework.
- Institutionally dependent sectors require relationship-specific investments and therefore rely on contract enforcement.
- Better institutions create comparative advantage in institutionally dependent goods.
- The North specializes in the institutionally dependent mixed good.
- The South may lose from trade because it loses the high-rent mixed-good sector.
- Labour in the South is particularly vulnerable; capital in the South tends to gain.
- Factor rewards may diverge after trade opening, contrary to standard factor-price-equalization intuition.
- The empirical test uses U.S. imports by country and four-digit SIC industry.
- Country institutional quality is measured by rule of law.
- Industry institutional dependence is proxied by dispersion of intermediate input use.
- Countries with better institutions capture higher U.S. import shares in more institutionally dependent industries.
- The result is robust to alternative institutional dependence measures, factor controls, financial development, distance, development, and alternative samples.

8 Conclusion

8.1 Core conclusion

Levchenko shows that institutions are a source of comparative advantage. Countries with better contract enforcement and property rights specialize in industries that require more complex production relationships. This conclusion is supported both theoretically and empirically.

8.2 Economic intuition

Institutionally intensive industries require many relationship-specific investments. When contracts are incomplete, these investments create holdup problems and factor-market distortions. Better institutions reduce these distortions. Under trade, the country with better institutions produces the institutionally dependent goods. This reallocates not only production but also rents.

8.3 Why the welfare result is important

The paper's most surprising result is that the South may lose from trade. This does not happen because trade destroys efficiency. It happens because trade reallocates the rent-paying institutionally dependent sector away from the South. The South gains from using the North's better institutions indirectly through imports, but loses the good jobs associated with the mixed sector. If the lost rents dominate the efficiency gains, the South loses.

8.4 Interpretation of the empirical evidence

The empirical results provide evidence of the institutional content of trade. Countries with higher rule of law have larger U.S. import shares in sectors with greater institutional dependence. This pattern remains after controlling for standard factor-proportions forces and many alternative explanations.

8.5 Limitations

The empirical measure of institutional dependence is a proxy, not a direct measure. Product complexity and intermediate input dispersion may capture other features of industries besides institutional reliance. The empirical design is cross-sectional, so it shows a robust pattern rather than a fully causal experiment. Also, institutions are multidimensional: contract enforcement, property rights, investor protection, political stability, and government expropriation may operate through different mechanisms.

8.6 Takeaway

The enduring takeaway is that institutions shape what countries produce and trade. Modelling institutions as incomplete-contracting frictions reveals welfare and distributional effects that a simple productivity view misses. Institutional quality is therefore not only a determinant of income levels; it is also a determinant of comparative advantage and the distribution of gains from trade.